CSCI 51.01: Operating Systems

“To err is human, to really @#$% things up requires a computer!”

Lecture Time!

Linking CSCI50 and CSCI51: Computers are much more than just their processors

What is an OS (or O/S or Operating System ~~or Overpriced Software~~)?

Before there were OS's: Guess who operated the system

An Interesting (Mis)Quote?

“I think that there is a world market for about five computers
- Thomas J. Watson (1943)

But did he make this quote? Or his son, in 1958?

Why make such a fuss over whether or not this particular person actually said this?

Prices Not Adjusted For Inflation

This supposed quote is actually plausible given the prohibitive costs of computers back then

- Renting an IBM 650 in the mid-1950s? That's at least $3,000 a month
- You could also buy it for $150,000

Why Study Operating Systems?

O/S may be software, but it:

- Consumes more resources than any other program
- Is the most complex program
- Is necessary for the use of the computer
- Is used by many users, for the reason above

Why Study Operating Systems?

Therefore, it needs to be EFFICIENT

Efficiency is measured through:

- Functionality
- Performance
  (Time and Utilization)
- Convenience and Cost

Mechanical Marie, episode 2

Functionality

The O/S has to control and allocate resources needed by other programs

Consider a play room with kids, toys,and a responsible adult acting as a guardian

Resources = Toys

Programs = Children

O/S = Guardian

Mechanical Marie, episode 10

Performance

The O/S should be fast in handling multiple jobs “at the same time” and/or in succession

Guardian must be able to swiftly and fairly resolve conflicts and distribute toys

Convenience and Cost

The O/S should make using the computer easier

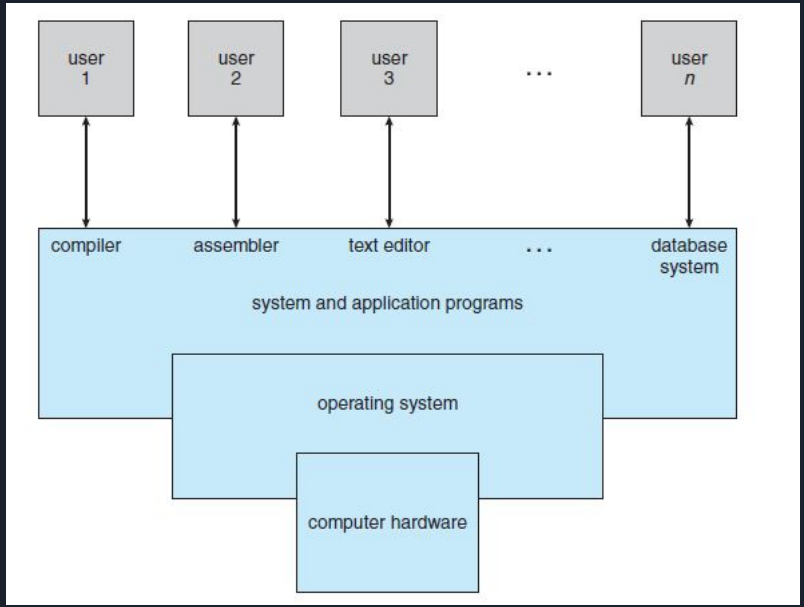
If something needs to be done, the O/S is there so you don't have to do it yourself

That's why people bring their kids to that play room – to make it “easier” to let their kids run/play

What is an O/S?

A layer of abstraction between hardware and software

- Virtual machine (think “translator”)
- A resource coordinator
- Reactive system (think "event-driven")



What is an O/S?

A collection of programs that integrate the hardware resources of the computer and make those resources available to the user in a productive, timely, and efficient manner

Operating Systems Ease the Pain By…

Performing the interface task with the hardware (file operations, memory paging, etc.)

- Would have to be done by the user if the O/S did not exist

Allowing a high-level user interface such as a GUI and/or a command line (CUI / CLI)

Operating Systems Ease the Pain By...

Being capable of multiuser and multitasking to utilize the hardware efficiently

- Otherwise, there's going to be a lot of waiting for earlier programs to finish

Making visible the "virtual" component of the system

Allowing program interaction, not only with the user, but with other programs

Why are Operating Systems Difficult to Create and Maintain?

Size

- An O/S is too big for one person or even a small group
- Current systems have millions of lines of code and involve 10-100 man years (not hours) to build

Why are Operating Systems Difficult to Create and Maintain?

Lifetime

- Operating systems remain longer than the programmers who originally wrote them
- As a result, as code is written and rewritten, the original intent is usually forgotten
  - Unix was designed to be a small and cute system, but now its many variants are intimidating to novices

Why are Operating Systems Difficult to Create and Maintain?

Complexity

The system must do difficult things: deal with I/O devices of varying speeds, multiplexing/juggling act, handle errors, etc.

~~If all else fails, it can always just crash~~

Multitasking

Must do several things “at the same time”

Why are Operating Systems Difficult to Create and Maintain?

General Purpose

Although it should be noted that there are operating systems that are especially made for specific machines

Such as airport control tower computers and video game consoles

https://www.reddit.com/r/itrunsdoom/comments/1ns5mk5/lawnmower_runs_doom/

A Brief History: Early 1950's, Mainframes

Early systems had no O/S

- Programmer is also the operator

Large machines run from a terminal; programs loaded through switches and card readers

- One program at a time

http://www.earlyofficemuseum.com

A Brief History: Early 1950's, Mainframes

Setup time was a problem, so programmers decided to hire an operator

Users would leave their programs with the operator, who would group these programs into batches with similar requirements in order to speed up processing

- Still one program at a time
- And the operator was probably really stressed out, so...

Batch Systems

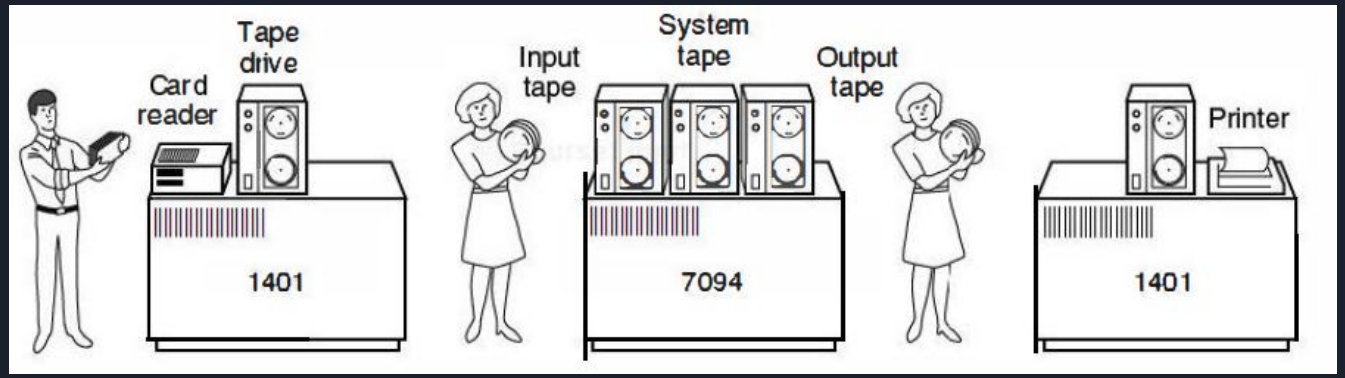
Simple Batch Systems were the first real O/S

- O/S was a simple program stored in one part of memory
- Programmers or operator puts cards in the card reader; no need to wait for current job to finish
- O/S would load one job at a time from the card reader into memory, output results somewhere (usually a printer) at the end of the current job, and transfer control to the next job; rinse and repeat

Batch Systems

Online Processing

Offline Processing*



Observation

However, even slow CPUs were idle most of the time

- It was NOT because the system “ran out of things to do”

Slow or not, they still worked in the microsecond range

- Electronic devices were (and still are) intrinsically faster than mechanical devices
- So why were the processors idle, even if there were still jobs lined up?

Offline Processing

Allowed jobs (programs) to be read ahead of time onto tape (which was faster than the card reader)

CPU could also write to tape instead of the slower printer

Programmer/Operator Behavior

Online Processing (Before): Go to computer, input from card reader, zzz, run program, output to printer, zzz, leave computer

Offline Processing (After): Copy from card reader to tape, zzz, go to computer, input from tape, z, run program, output to tape, z, leave computer, output to printer, zzz

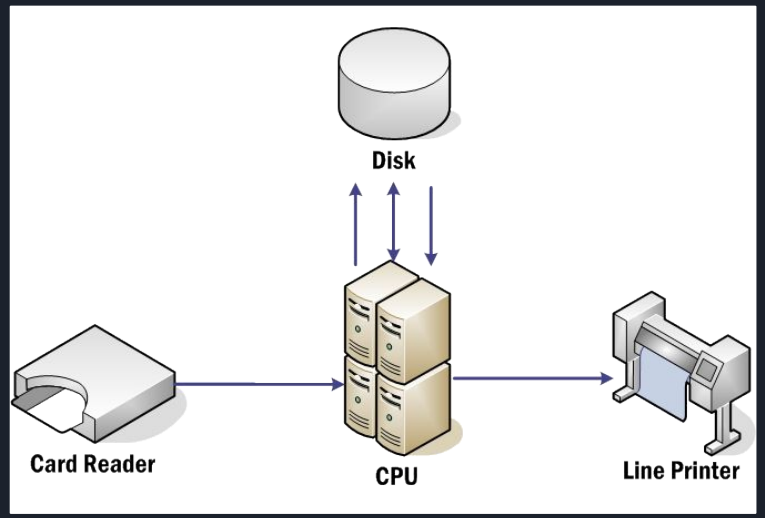
Note: Tapes were faster than punched cards

Disks and Spooling

Spool: Simultaneous Peripheral Operation On-Line

Spooling allowed jobs to be read ahead of time onto disk

- Disk was used as a huge system buffer/cache
- Disks are MUCH faster than tapes



Disks and Spooling

Instead of "driving" the printer themselves when needed, programs can now just pass their print requests to a queue

- The queue being on the disk, of course

Then, a spooler program is responsible for looking at that queue and driving the printer

- Spooling is actually a form of multiprogramming

Program Behavior

No Spooling (Before): Compute stuff, request print-out, zzz, compute more stuff, request another print-out, zzz

With Spooling (After): Compute stuff, request print-out, compute more stuff while waiting for first print-out, request another print-out, zzz

Disks and Spooling

In other words:

- Processors were fast enough (for the programs then, anyway)
- But long wait times were being caused by slow I/O, and the computer couldn’t do anything else while waiting
- We had to think of ways to allow the processor to continue doing other things while waiting

Need For Speed

"Necessity is the mother of invention"

From here on, O/S designs would reflect what everyone needs: faster computers

Up next: O/S types